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Turnout Maintenance in Heavy Haul Environments and Beyond

Richard Stock - Global Head of Rail Solutions

Ash Athukoralalage - Wheel/Rail Interface Engineer

DALLAS, TX

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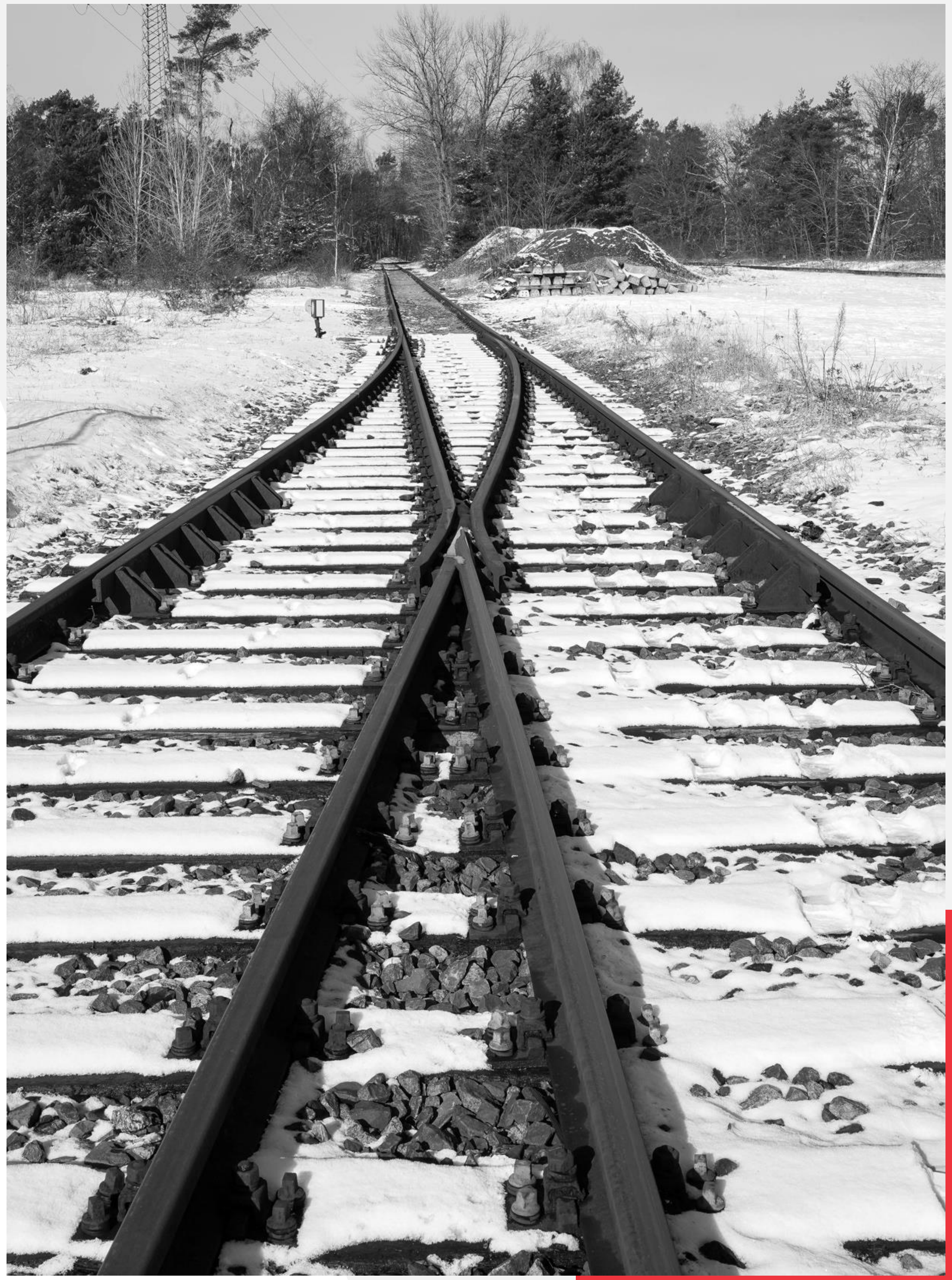
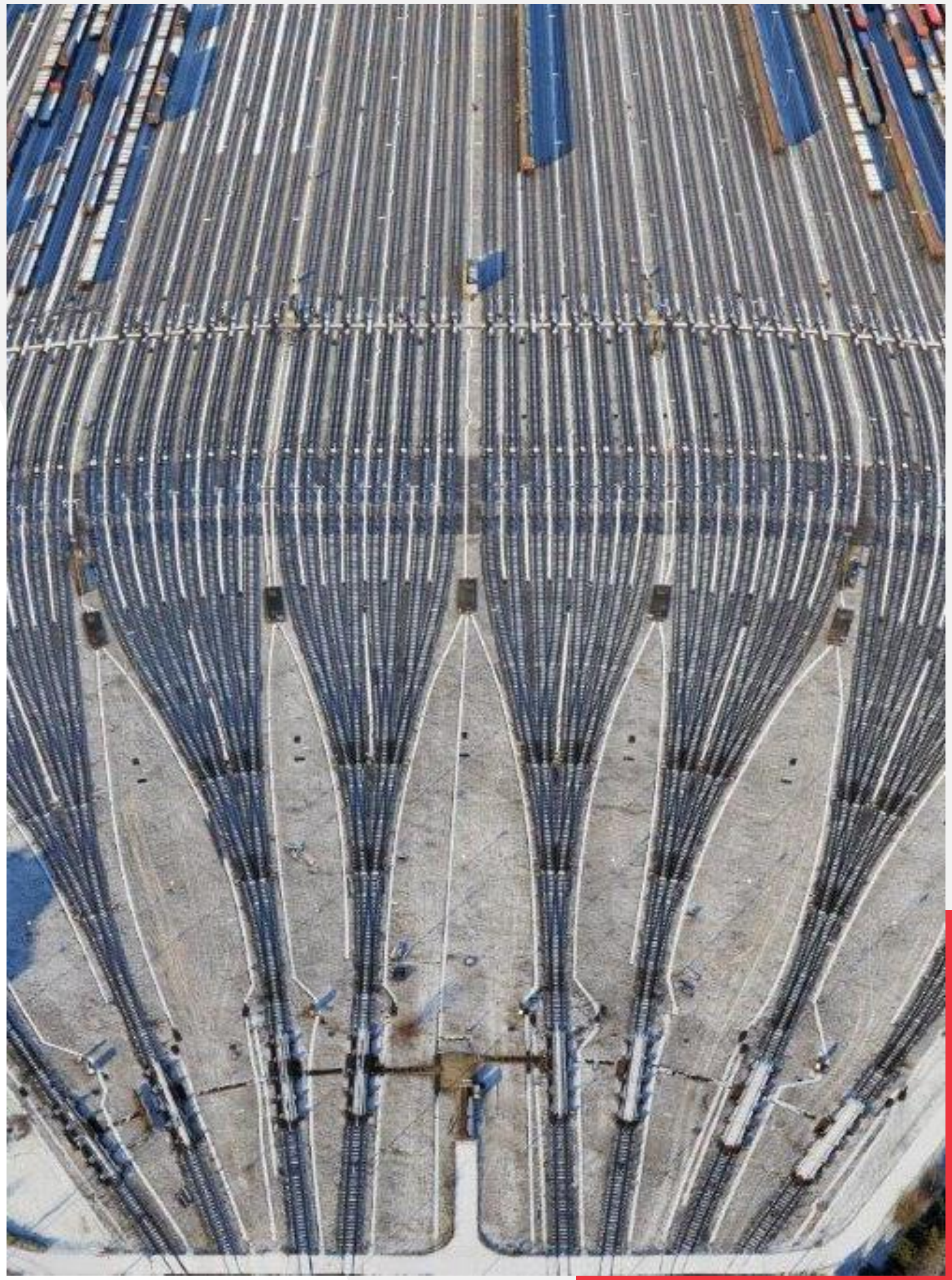
Function / Definition of a Turnout

- The purpose of a turnout is to permit engines and cars to pass from one track to another (AREMA)
- Layout permitting the passage of rolling stock between two tracks and one common track (EN13232)



Turnout Statistics for Class 1 - I

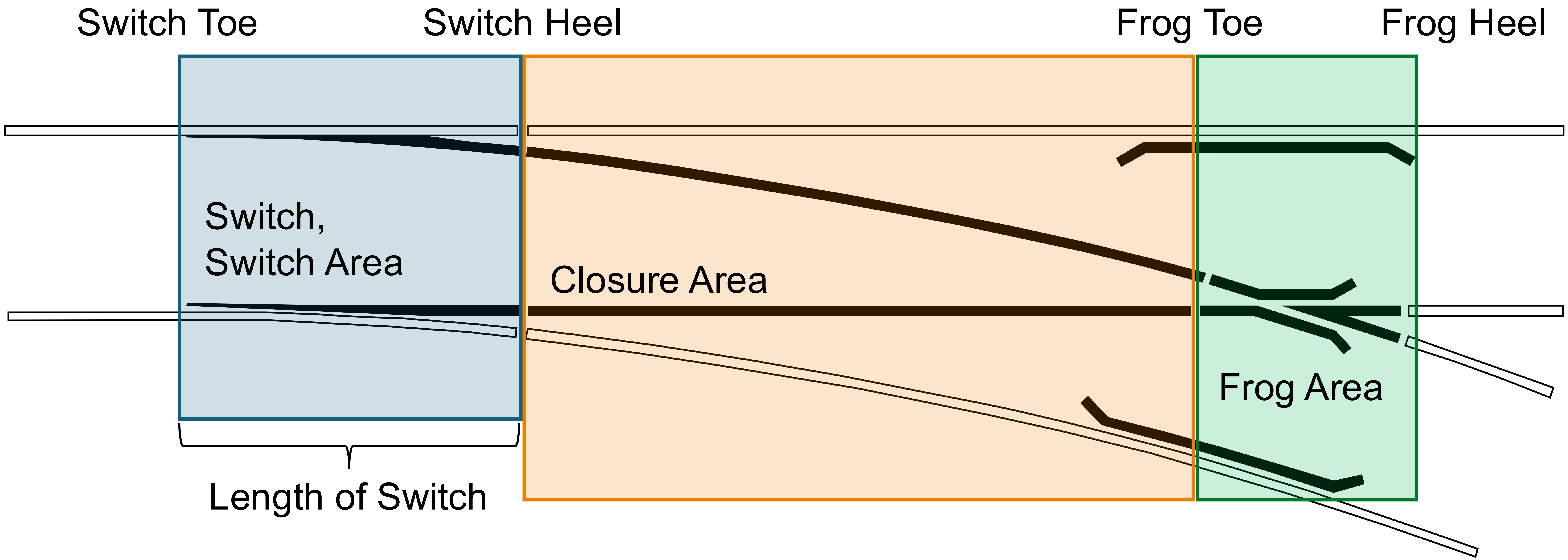
- No Class 1 statistics available (not reported) on number of turnouts
- Two datapoints
  - UP in 2013: 31500 turnouts
  - CN in 2014: 20000 turnouts
- Total # scaled by route miles (1-1.5):
  - Approx. 142000-213000 turnouts
- Germany: 65000 turnouts (20700mi)


Marshalling Yard Hamburg Maschen, Aufwind Luftbilder

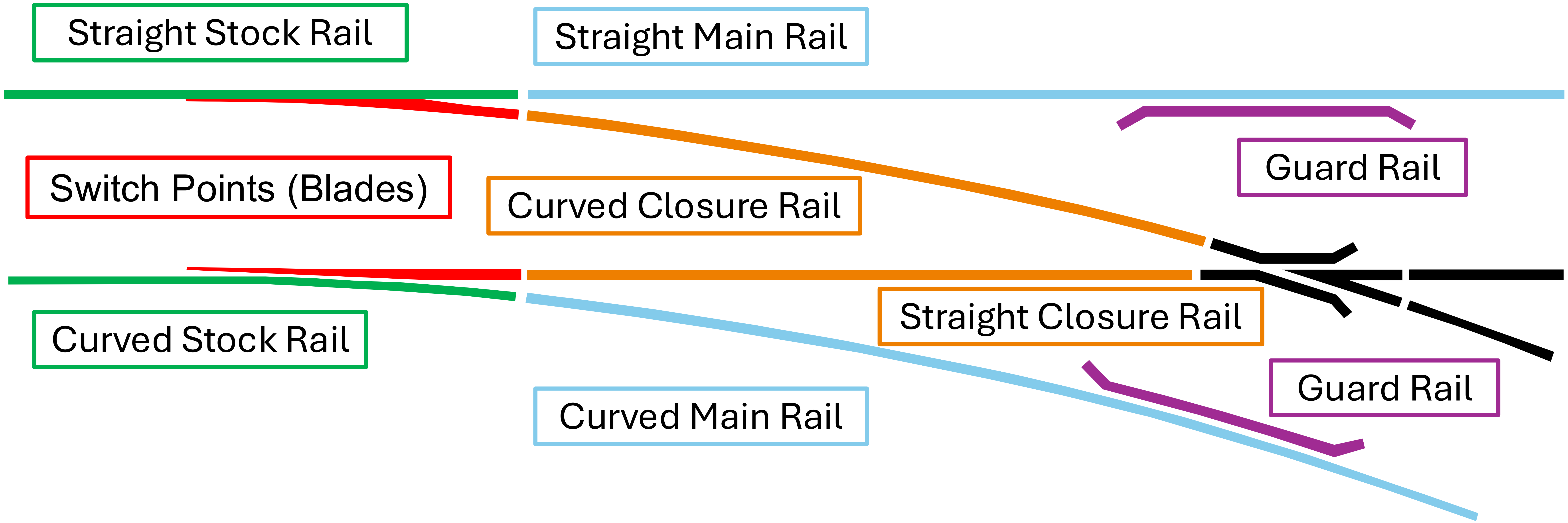
Costs for a Turnout

- Even more difficult to get meaningful data
- Turnout type/material cost, labour cost, operational hinderance cost
- Range of 50k$ – 500k$ per turnout
- Most expensive track component
- Premature failure very expensive

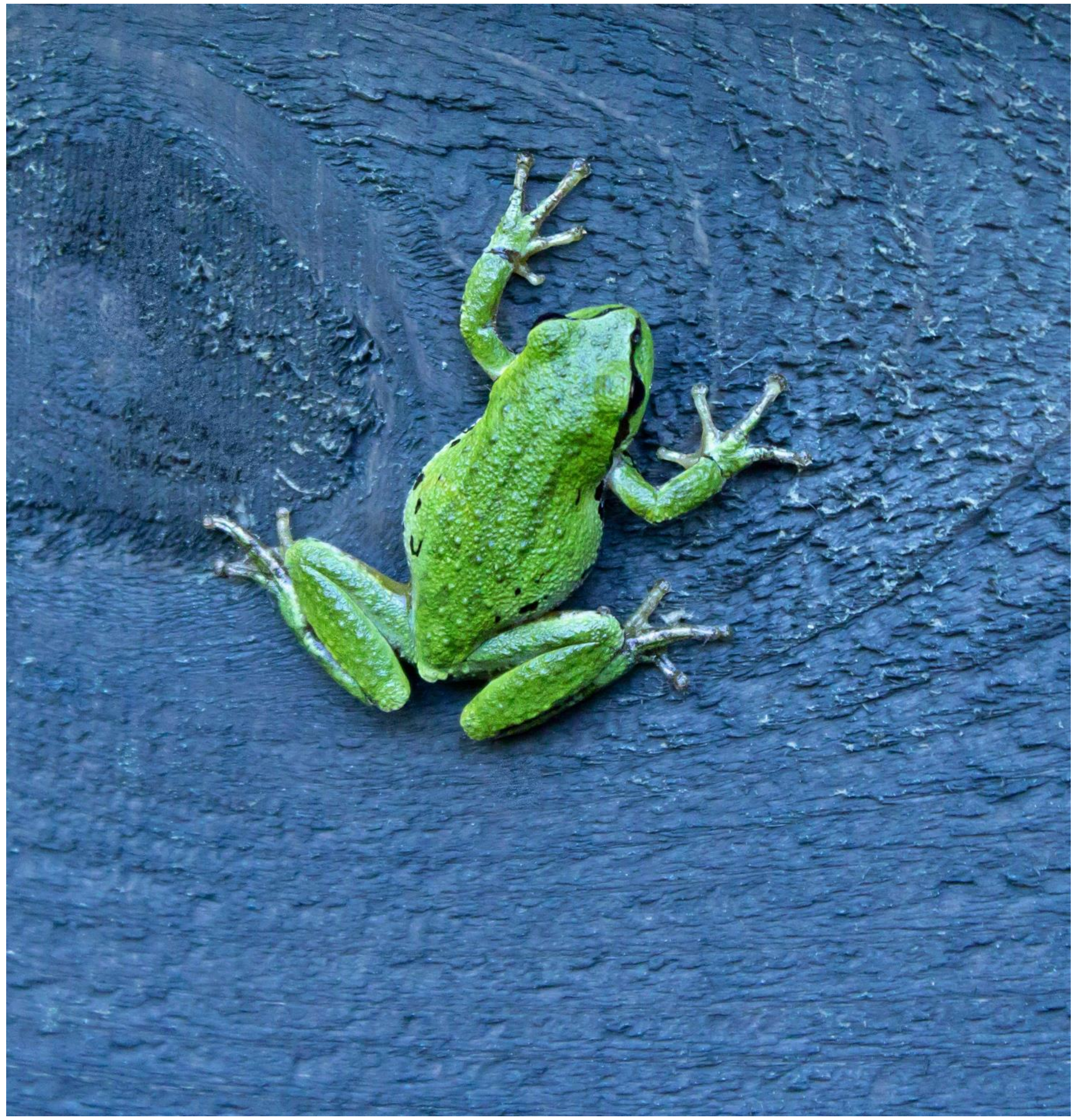
Turnout Layout



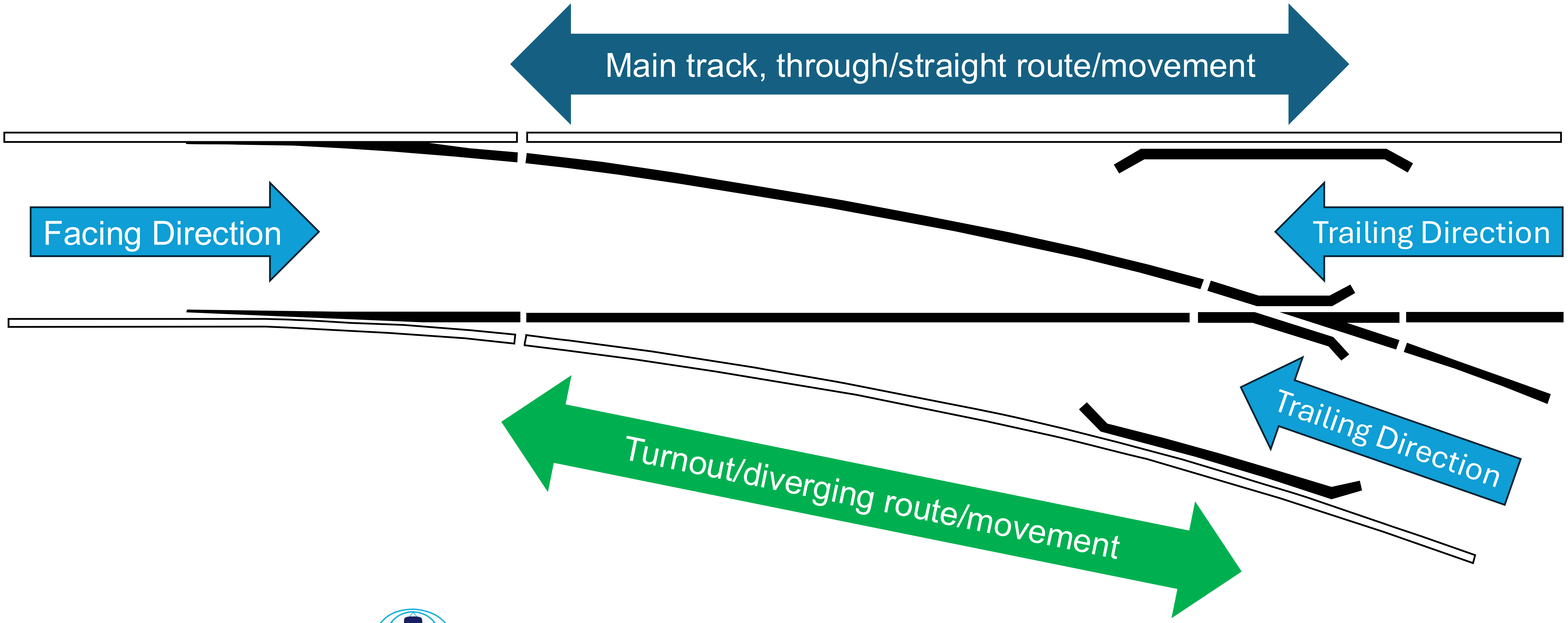
Turnout Components



Frog



Turnout Specifics

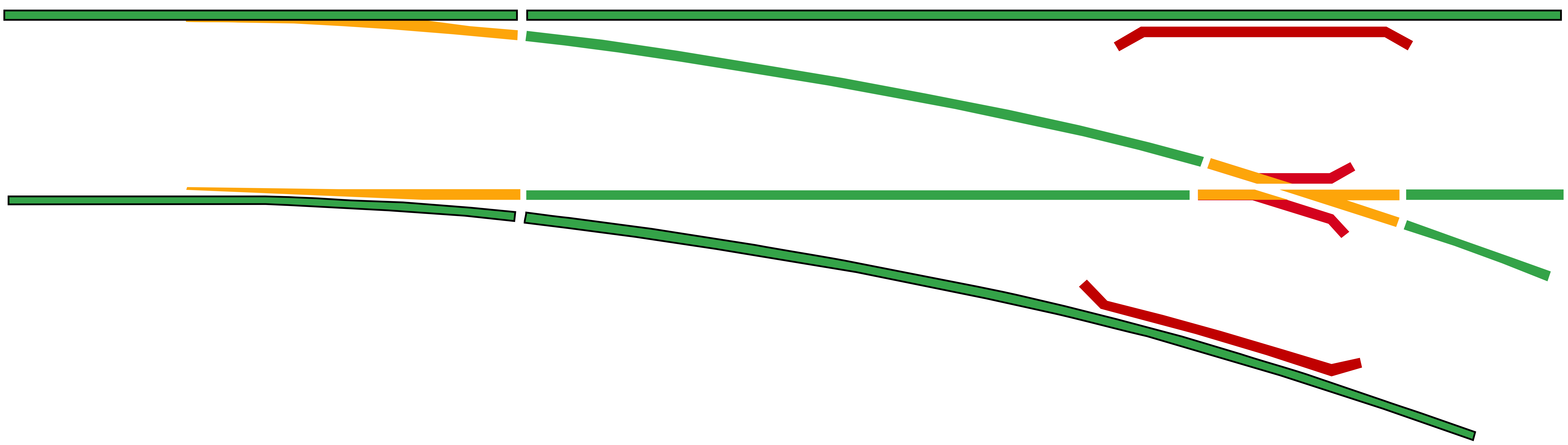


Challenges for Turnout Treatment

Full Profile

Partial Profile

Limited Clearance

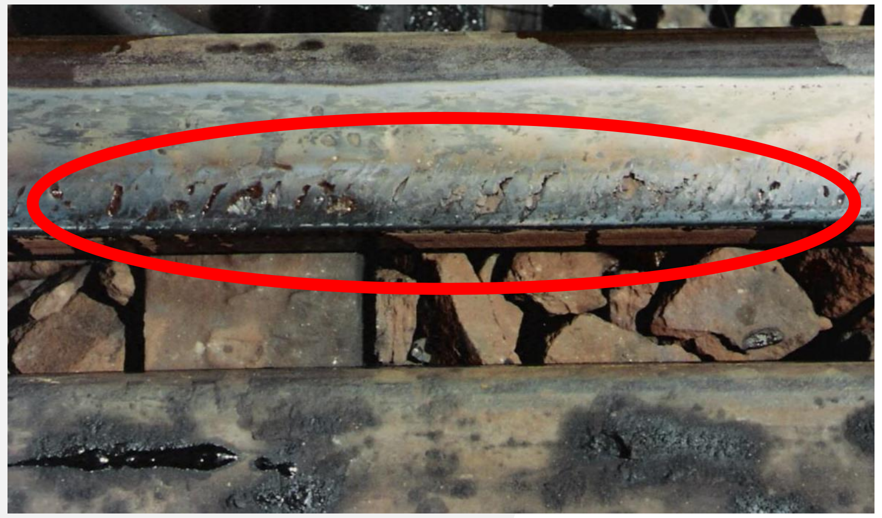


Turnout Damage

- Remove surface damage
  - Rolling Contact Fatigue (RCF), Spalling
  - Squats / Studs
- Restore railhead profile
  - Reduced contact stresses – less damage
  - Shift contact from critical stress areas
  - Reduced dynamic forces
- Reduced plastic flow
  - Live extension of blades and frogs

Significant switch rail RCF

Field side defects on switch blade

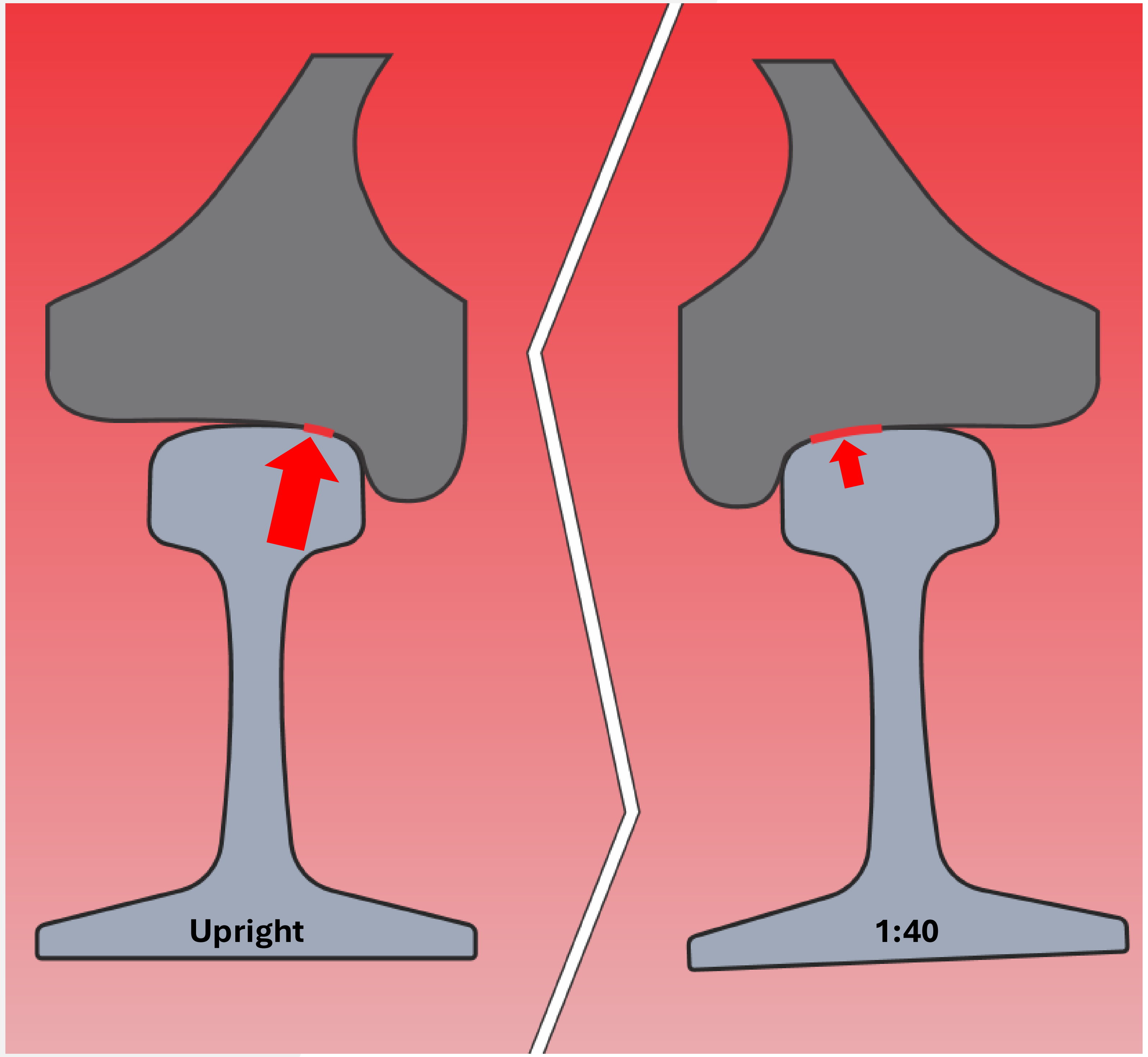


Gauge Corner Cracks

Gauge corner “lipping” and switch blade damage

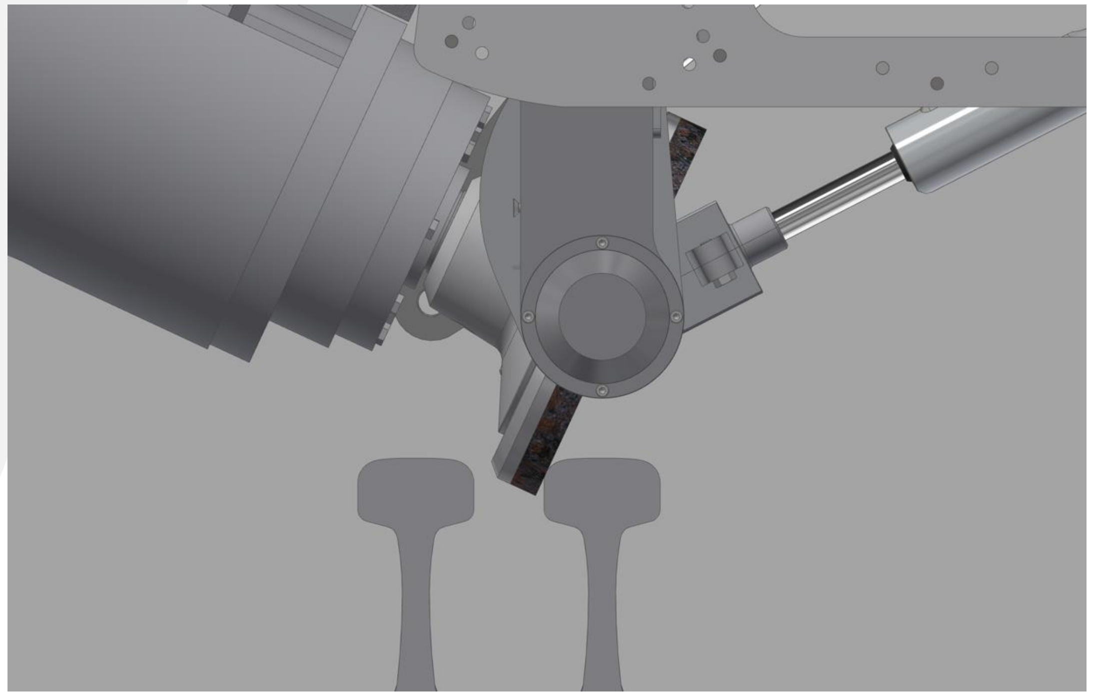
Correction of Rail Geometry

- Regular track: 1:40 (or 1:20) rail cant
- Upright rail installation in turnout
  - Dependent on turnout design: several transitions per turnout – dynamic effects
- Upright rail and wheel profile create narrow, high stress contact conditions
  - Increased RCF development
- Correction of rail geometry with treatment
  - 1:40 profile throughout turnout and subsequent delayed damage formation

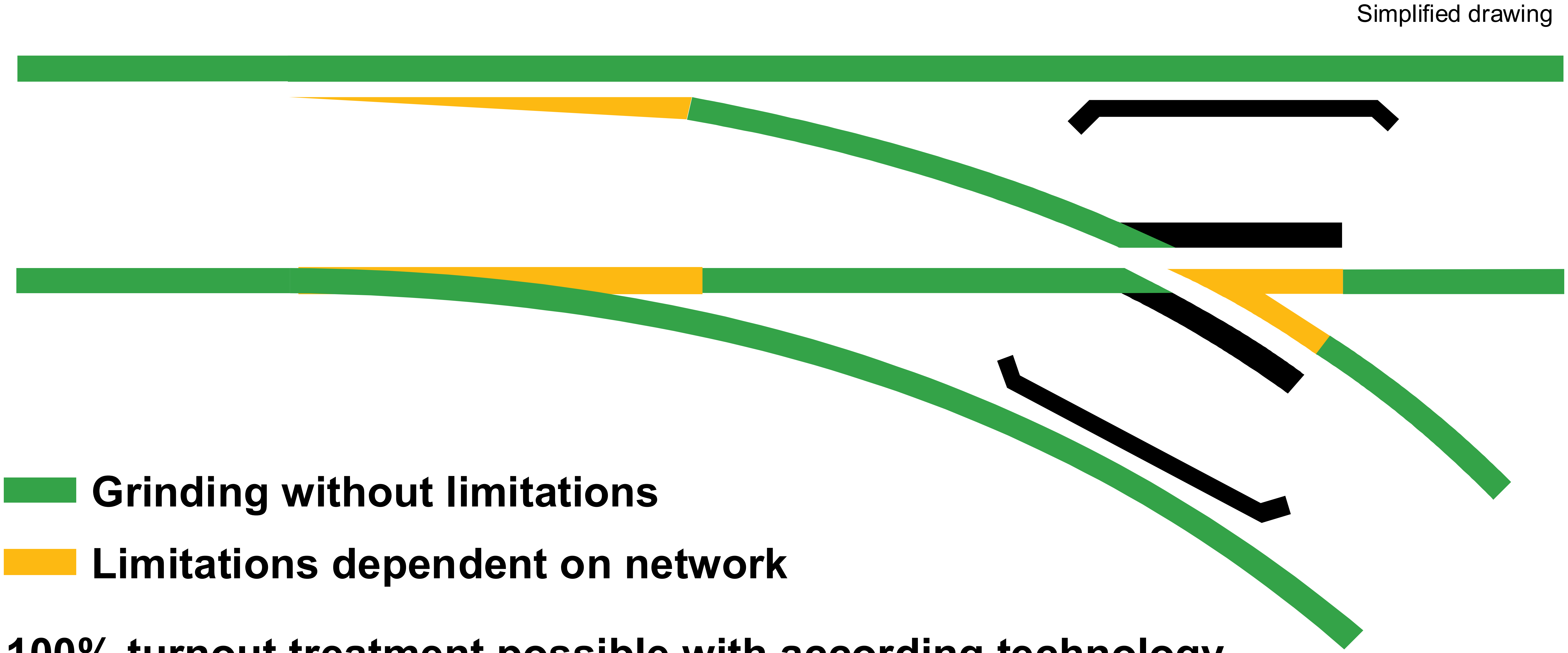


Grinding Technology

- Specialty grinders with 12-24 stones
  - Rail-bound or hi-rail – compact design
- Partial profiles and limited clearance
  - Specific designed grinding stones – dinner plate stones
  - Independent motor operation, variable angles, and selective lifting
  - Lateral motor offset to reach around obstacles, wing rails, and other angular constraints



Covered Areas by Grinding



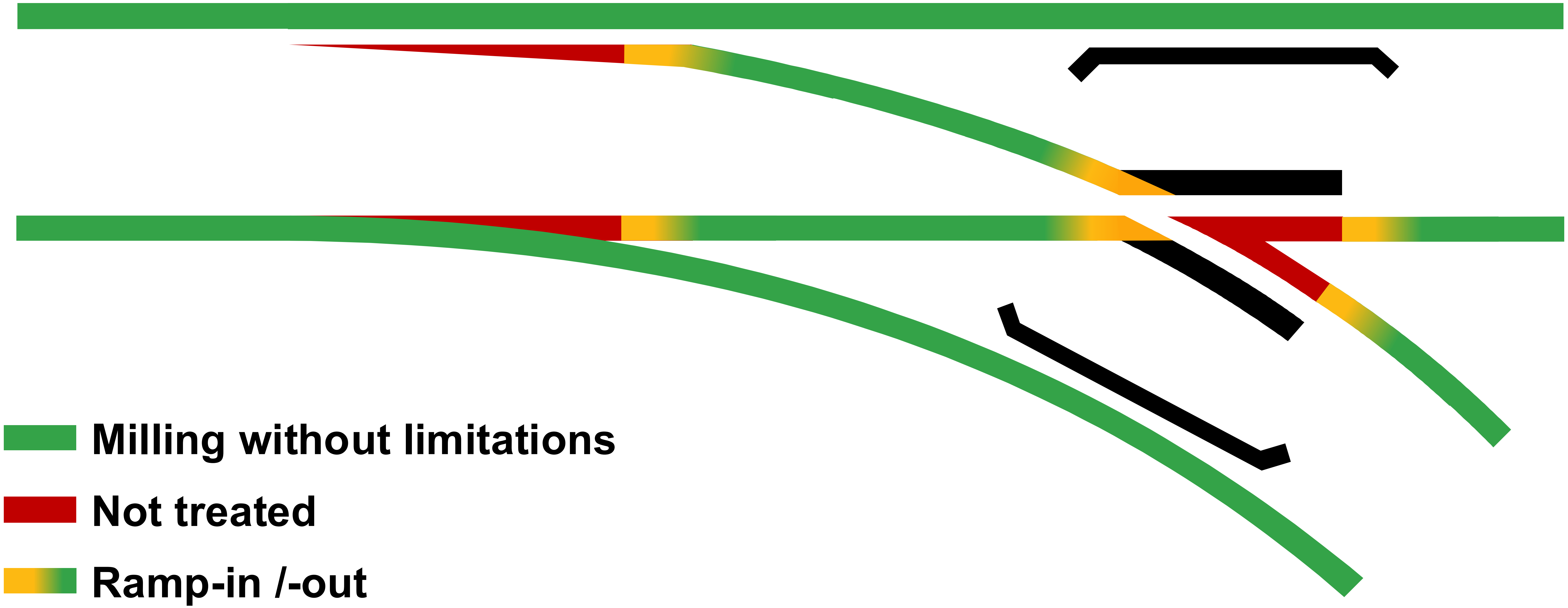
100% turnout treatment possible with according technology

Milling Technology

- Every milling machine could treat turnouts
  - Compact milling machines more suitable (hi-rail or rail-bound)
  - Software and minor hardware adjustment (sensors)
- Partial profiles and limited clearance
  - Some turnout parts cannot be treated

Covered Areas by Milling

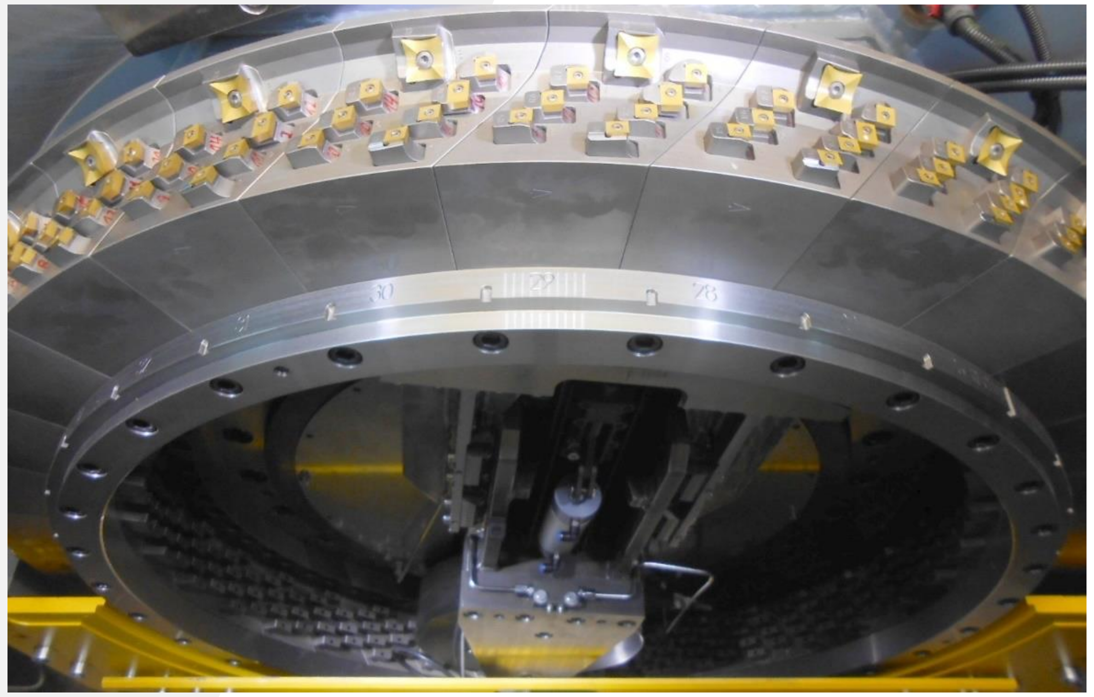
Simplified drawing

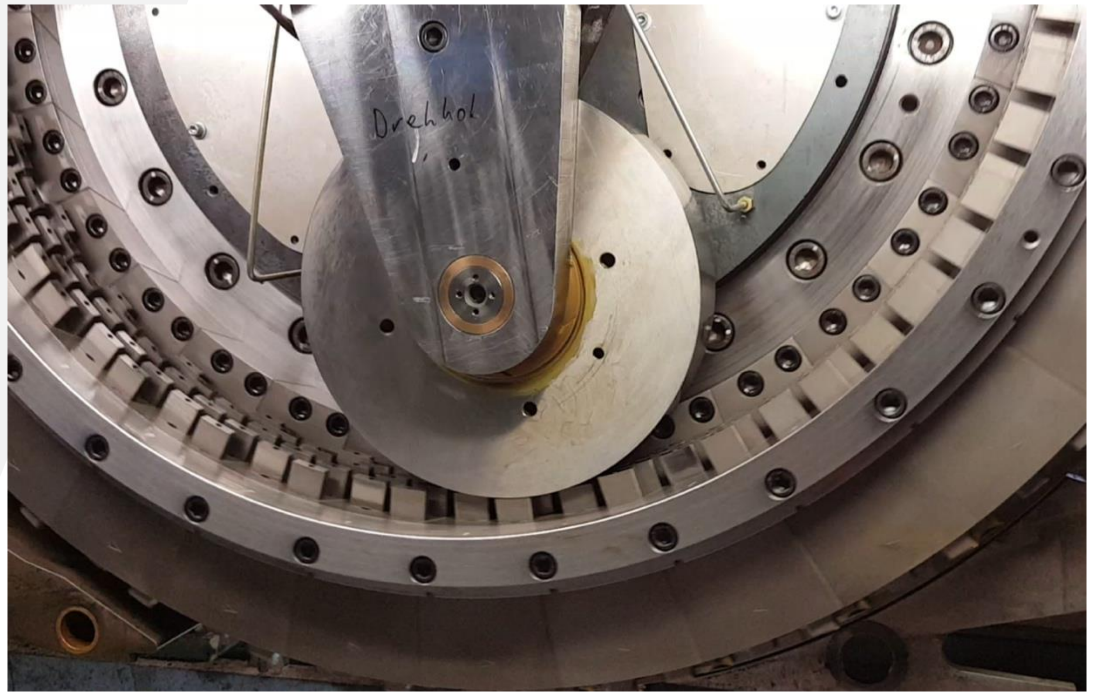


- Milling without limitations
- Not treated
- Ramp-in /-out

Rotational Planing Process

- Combination of milling and planing process
- Milling inserts move parallel to rail surface, when in contact with rail
  - Spring loaded inserts
  - Smooth surface finish
- Possibility to disengage specific insert-traces
  - Partial profile



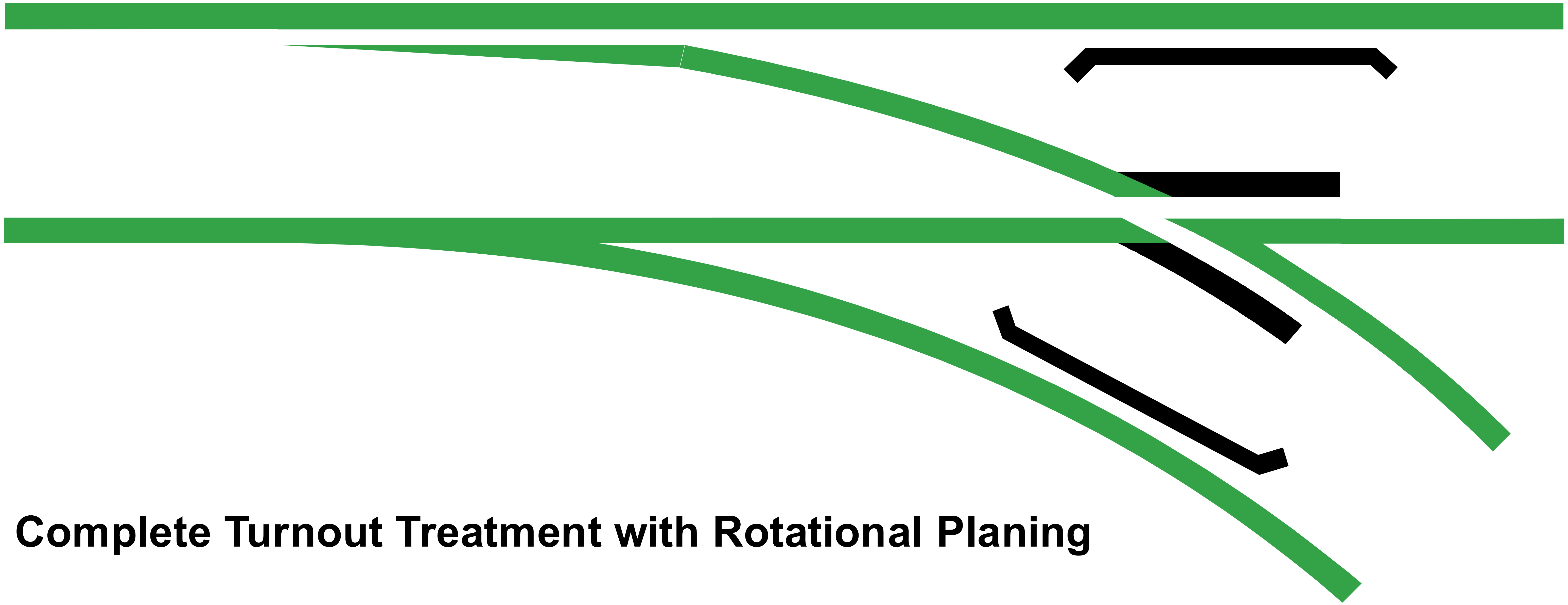


Rotational Planing Technology

- Dedicated, compact machine (D-HOB)
  - 3 piece consist
  - Control car, planing car, chip bunker and oscillating grinding unit
- Partial profiles and limited clearance
  - Profile adjustment during process
- 100% of turnout can be treated

Covered Areas by Rotational Planing

Simplified drawing



Complete Turnout Treatment with Rotational Planing

Application Example: Australia

- Australian HH networks
  - Up to 40t axle load and 200-350 MGT/a
- Major operators (ARTC, BHP, Rio Tinto, FMG)
- Cyclic turnout grinding every 25-30 MGT
- General damage removal
- Correction of vertical rail installation
- Frogs and switch points as high-impact zones

Turnout Grinding in Australia: Operations

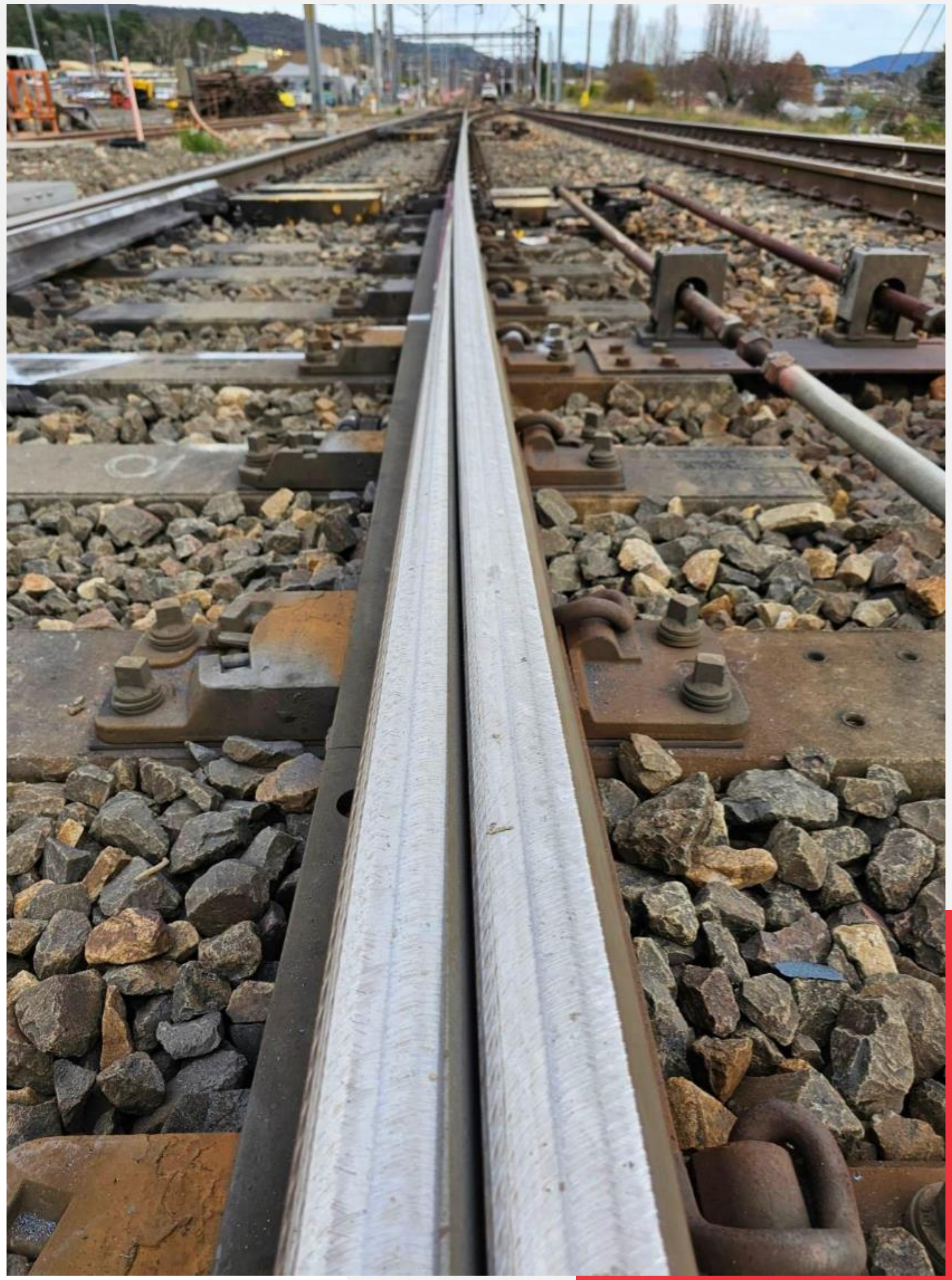
- Turnout inspection and preparation
  - Mark turnout including lift and drop-off locations
  - Determine any turnout issues
- Typical preventive grind time including pre-inspection and quality control (0.2mm metal removal, ½ turnout): 15 min
  - Full turnout: 25 min
- Corrective grind full turnout: 45-60 min

Return on Investment

- Renewal deferral saves $350k – $700k (USD) per turnout
- ROI consistently above 100% in heavy-haul networks
  - Biggest ROI from avoided derailments and deferred renewals
  - Highest in Pilbara and CQCN (Central Queensland Coal Network) due to extreme tonnage/axle loads

Australia: Life Extension of Turnouts

- High tonnage – low axle load turnouts
  - Without grinding: 5-10 years
  - With grinding: 20+ years
- High Tonnage – high axle load turnouts
  - Without grinding: 2-5 years
  - With grinding: 10-15 years
  - Change from turnout replacement to component replacement and life extension



A Message from the Presenter

Benefits of Turnout Grinding: Austria

- Austrian Rail Network:
  - 22.5t axle load, mixed traffic, 3100 mi
- Comparing scenarios of no grinding vs. optimized grinding (complete turnout)
- Service life extension from initially 20 to 30 years
  - Service life has most important influence on total costs
- Overall calculated savings of up to 37%
  - Cost savings of 230k to 350k USD per turnout
- If in doubt, grinding more frequently is often more economical (than doing nothing)
  - Optimize frequency later

Germany: Rotational Planing (D-HOB)

- Turnout inspection and preparation
  - Mark turnout for full/partial profile sections
  - Determine any turnout issues
- Usually 1 pass per turnout direction
- Quality control (integrated measurement technology)
- Process control by external operator
- Total time can vary – faster than grinding for metal removal ≥ 1mm

Germany: Application Example D-HOB

- Preventive and corrective turnout treatment
  - 0.2 – 2mm metal removal per pass
  - Full damage removal
  - High quality surface finish
- No turnout cleaning and no fire prevention measures required
- Used on conventional and high-speed turnouts (≤ 174mph through / ≤ 125mph diverging)
- Approx. 500-800 shifts per year (vs. 250-600 shifts for switch grinding)

Turnout Milling

- Turnout inspection and preparation
  - Mark turnout concerning milling tool lift and drop-down locations
  - Determine any turnout issues
  - Milling machine preparation
- Reduced milling speed
- Process control by external operator

Example HH Milling: Australia

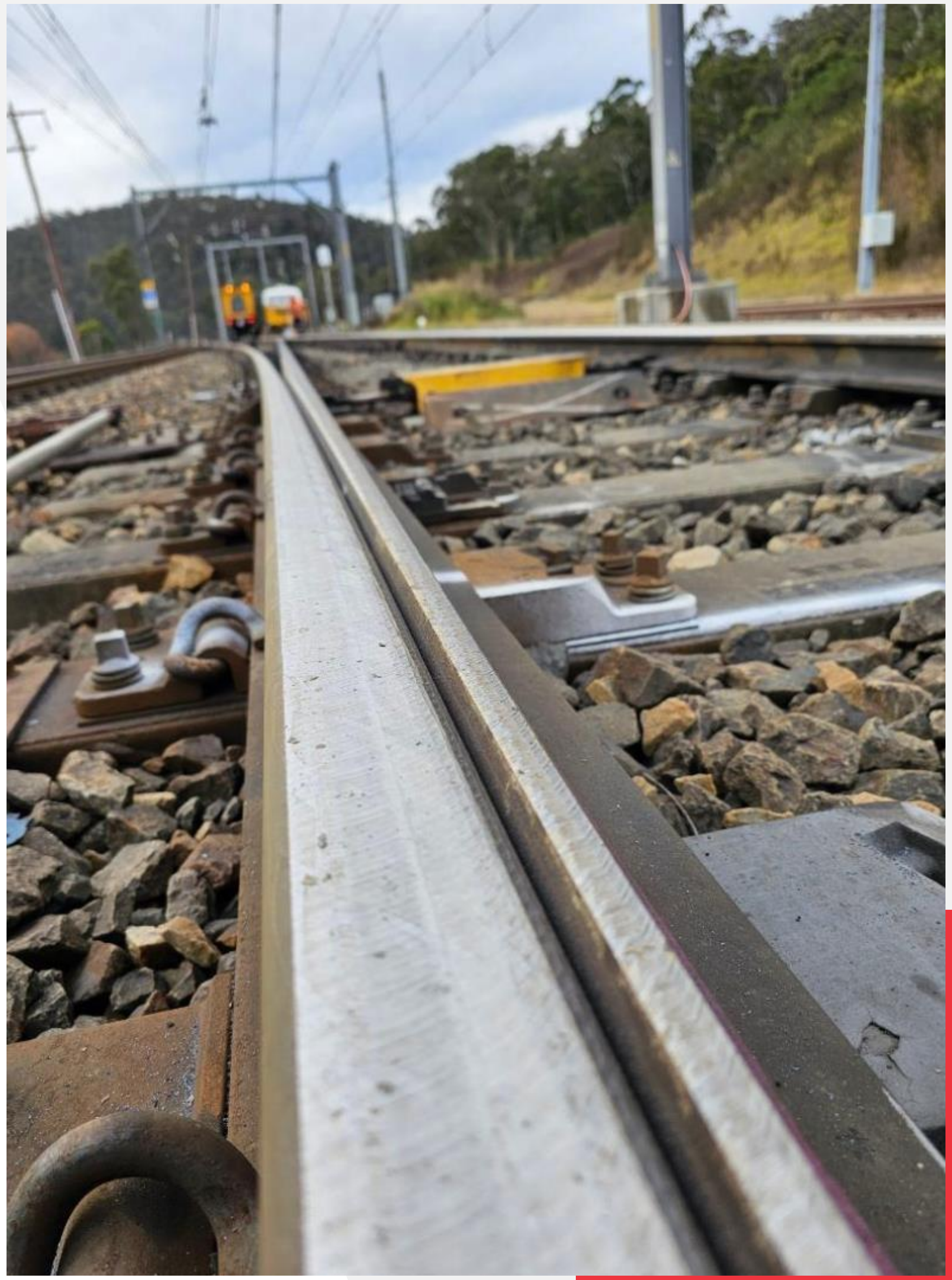
- Heavy defect removal on HH turnouts
  - Up to 2mm
  - 1-3 passes (machine dependent)
- Spark and dust free treatment
- Complimentary to rail grinding
  - Time efficiency for heavy metal removal
- Untreated areas: hand grinding
- Average time (1-pass): 45-60 min per full turnout (excl. hand grinding)

Measurement Technology

- FRA specifies minimum visual / manual inspection requirements
- Advanced turnout inspection for comprehensive turnout condition
  - Look-up inspection with built in sensors
    - Sound, displacement, vibration, …
  - Vehicle based look-down inspection
    - Vision systems with AI pattern recognition, laser measurement systems, electromagnetic systems, UT, …

Summary

- Turnout as one of the most expensive track components
- Turnout damage similar to open track
- Benefits of turnout treatment:
    - Life extension
    - Avoided derailments
- Technologies: Grinding, Rotational Planing, Milling
- Measurement Technology



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Thank you for your attention!

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